

QA Evidence — All ~/api_keys.sh Providers + Positive-Codebase-Answer + Real LLMsVerifier Consumption

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QA Evidence: All ~/api_keys.sh Providers, Positive-Codebase-Answer Guarantee, and Real LLMsVerifier Consumption

Date: 2026-06-14 **Scope:** HelixCode (helix_code/) + HelixAgent + LLMsVerifier submodules **Commit under test:** a99a5e9d — *feat(tui, llm, verifier): use all ~/api_keys.sh providers + guarantee positive codebase answer + real LLMsVerifier consumption* **Anti-bluff posture (CONST-035 / Article XI §11.9):** This document states only what was observed in this session. Where a thing was NOT proven, it is explicitly flagged as such. **No API key value is printed anywhere in this document.**

1. Verdict

PASS (with honest limitations). Three independently-verified outcomes:

1. **Provider breadth** — HelixCode's TUI live provider count went **5 → 18**, with 8 newly-added OpenAI-compatible providers registering live models at startup and the remaining unreachable ones gracefully skipping (no crash, no fake entries).
2. **Positive-codebase-answer guarantee** — proven live: the agent issues a real tool call before any codebase claim and answered affirmatively with the real `git_status` file list (captured frame).
3. **Real LLMsVerifier consumption (CONST-036)** — proven: HelixCode in remote mode received **all 78 real models** populated in the verifier DB, not the 7 embedded fallback entries.

The following are explicitly **NOT** claimed as working and are documented in §7 (Honest Limitations): several free-tier keys are invalid/suspended; one provider hit a rate limit on a heavy prompt; one HelixAgent provider has no chat-entitled model yet; one key was excluded entirely.

2. What Changed

2.1 HelixCode (helix_code/)

- **internal/llm/openai_compatible_catalogue.go** (new, +254 lines) — a **data-driven catalogue** of hosted OpenAI-compatible providers. Base URLs are sourced from HelixAgent’s verified provider table (not hand-invented). Each entry is keyed off an env var; a provider only activates when its key is present.
- **GetType() collision fix** — GetType() now derives a distinct type per catalogue provider, fixing a collision that previously conflated multiple OpenAI-compatible providers.
- **applications/terminal_ui/env_providers.go** (new, +20 lines) — a **registration loop** that registers every present-key catalogue provider into the TUI provider manager at startup.
- **applications/terminal_ui/main.go** (+48 lines) — the positive-codebase-answer changes (see §4): system-prompt instruction requiring a tool call before any codebase claim, resolveRepoRoot pinning git_status to the git root, and ensemble-first picker behaviour.
- Tests added alongside: openai_compatible_catalogue_test.go (+224), env_providers_test.go (+11).

2.2 HelixAgent (submodule)

- Builds and boots.
- From the same keyset, discovered a **25-provider / 993-model catalogue**.
- **6 providers live-verified**: cerebras, codestral, deepseek, groq, mistral, openrouter.
- A new **INFERENCE** provider was added but its account currently has no chat-entitled model (see §7).

2.3 LLMsVerifier (submodule)

- Builds and serves on **:8095**.
 - DB populated with **78 real models**: groq 16, deepseek 4, mistral 58.
 - The HelixCode verifier client was **reconciled** to the verifier’s actual contract: envelope shape, /api/providers endpoint, Unix timestamp handling, and numeric id + model_id fields.
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3. Provider-Support Matrix

Live model counts below are from the HelixCode TUI startup log captured this session. “Wired in HelixCode” = registered via the catalogue/registration loop at a99a5e9d.

Provider	Key present	Wired in HelixCode	Live models (startup)	Status
cerebras	yes	yes	2	Registered live. Hit 429 RPM on heavy multi-turn prompt (see §7).
hyperbolic	yes	yes	5	Registered live.
novita	yes	yes	138	Registered live.
sambanova	yes	yes	6	Registered live.
nvidia	yes	yes	121	Registered live.
zai	yes	yes	7	Registered live.
venice	yes	yes	90	Registered live.
chutes	yes	yes	13	Registered live.
siliconflow	yes	yes	69	Fixed this session: base URL .cn → .com. Now registers 69 models.
fireworks	yes	yes	0	Gracefully skipped (account suspended — 401/403/404/412 class). Documented key-side.
kimi	yes	yes	0	Gracefully skipped (invalid key). Documented key-side.
upstage	yes	excluded	—	Excluded: no / models endpoint to enumerate.
codestral	yes	excluded	—	Excluded: no / models endpoint to enumerate. (Live-verified separately in HelixAgent.)
tencent	invalid	no	—	

Provider	Key present	Wired in HelixCode	Live models (startup)	Status
				TENCENT_CLOUD key invalid — not added.

Notes. - The 8 providers cerebras / hyperbolic / novita / sambanova / nvidia / zai / venice / chutes are the ones whose live registration drove the 5 → 18 count. - fireworks and kimi are *wired* but return 0 live models because their keys are unusable; they skip gracefully rather than emitting placeholder entries (anti-bluff: no fabricated models). - upstage and codestral were excluded from the catalogue because they expose no `/models` listing endpoint; this is a deliberate exclusion, not a silent failure.

4. Positive-Codebase-Answer Proof

What was built. Three coordinated changes in `applications/terminal_ui/main.go`: 1. **System prompt** (`buildToolLoopSystemPrompt`) instructs the agent it operates INSIDE the repository and must **make a tool call before any codebase claim**, then confirm concretely. 2. **resolveRepoRoot** pins the `git_status` tool to the actual git root, so the agent reads real repository state. 3. **Picker is ensemble-first**, so a capable model handles the codebase question.

Live proof (captured frame). Using cerebras `gpt-oss-120b`, the agent answered:

“I can see your repository... So yes—I have direct read access to the codebase and its git state”

and rendered the **real `git_status` file list** in the same response. The affirmative claim was backed by an actual tool call returning real repository state — not a hardcoded or simulated answer.

Honest scope of this proof. This was verified for the cerebras `gpt-oss-120b` path in a single captured frame. It demonstrates the mechanism works end-to-end; it is not a claim that every one of the 18 providers was individually exercised through the same flow.

5. Real-Verifier-Consumption Proof (CONST-036)

Claim under test. HelixCode consumes LLMsVerifier as the single source of truth, not embedded fallback data.

Setup. LLMsVerifier built, served on `:8095`, DB populated with **78 real models** (groq 16, deepseek 4, mistral 58). HelixCode run in **remote mode** against the verifier.

Result. HelixCode received **all 78 models** from the verifier — **not the 7 embedded fallback** entries it would use when the verifier is unreachable. Receiving 78 (the live DB count) rather than 7 (the static fallback count) is the discriminating evidence that real verifier consumption occurred.

Reconciliation work that made this real. The HelixCode verifier client was aligned to the verifier's actual API contract: response envelope shape, the `/api/providers` endpoint, Unix-timestamp parsing, and numeric `id + model_id` fields. Before reconciliation the client did not correctly parse the verifier's real responses.

6. Commit Reference

All of the above landed at commit **a99a5e9d**:

```
feat(tui,llm,verifier): use all ~/api_keys.sh providers +  
guarantee positive codebase answer + real LLMsVerifier  
consumption
```

Touched (per `git show --stat`): `internal/llm/openai_compatible_catalogue.go` (+254), `internal/llm/openai_compatible_catalogue_test.go` (+224), `applications/terminal_ui/env_providers.go` (+20), `applications/terminal_ui/env_providers_test.go` (+11), `applications/terminal_ui/main.go` (+48).

7. Honest Limitations (anti-bluff)

These are real, in-session findings. None is hidden behind a green summary line.

1. **Invalid / suspended free-tier keys.** fireworks (account suspended), kimi (invalid key), upstage, and tencent keys are not usable. fireworks and kimi are wired but contribute 0 live models and skip gracefully; tencent (TENCENT_CLOUD) was invalid and **not added**.
 2. **Rate limit on heavy prompt.** The cerebras free tier hit a **429 RPM limit** on the heavy multi-turn AGENTS.md prompt. The provider is registered and works for normal prompts; the limit is a free-tier throughput ceiling, not a wiring defect.
 3. **HelixAgent INFERENCE provider has no chat-entitled model yet.** The INFERENCE provider was added to HelixAgent, but its account currently exposes no chat-entitled model, so it cannot yet serve chat completions.
 4. **upstage / codestral excluded from the HelixCode catalogue.** Reason: neither exposes a `/models` enumeration endpoint. codestral itself IS live-verified inside HelixAgent (one of the 6), so this is a HelixCode-catalogue listing limitation, not a provider-dead finding.
 5. **Coverage scope of the positive-answer proof.** Proven for the cerebras gpt-oss-120b path (§4). Not every one of the 18 wired providers was individually driven through the codebase-answer flow this session.
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8. Reproduction Pointers

- Catalogue definition: `helix_code/internal/llm/openai_compatible_catalogue.go`
- TUI registration loop: `helix_code/applications/terminal_ui/env_providers.go`
- Positive-answer prompt + repo-root pin: `helix_code/applications/terminal_ui/main.go` (`buildToolLoopSystemPrompt`, `resolveRepoRoot`)
- Catalogue unit tests: `helix_code/internal/llm/openai_compatible_catalogue_test.go`
- Verifier endpoint observed: `http://localhost:8095/api/providers`